

Q1

Vertex is at $(\pm 6, 0)$

$$\therefore a = 6$$

Let the hyperbola is $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

Putting point $P(10, 16)$ on the hyperbola

$$\frac{100}{36} - \frac{256}{b^2} = 1 \Rightarrow b^2 = 144$$

$$\therefore \text{hyperbola is } \frac{x^2}{36} - \frac{y^2}{144} = 1$$

$$\therefore \text{equation of normal is } \frac{a^2 x}{x_1} + \frac{b^2 y}{y_1} = a^2 + b^2$$

$$\therefore \text{putting we get } 2x + 5y = 100$$

Q2

$$\text{Hyperbola : } \frac{x^2}{4} - \frac{y^2}{2} = 1$$

So, let the point on it be $(2 \sec \theta, \sqrt{2} \tan \theta) = (x_1, y_1)$

$$\text{So, equation of tangent} = \frac{2 \times \sec \theta}{4} - \frac{\sqrt{2} y \tan \theta}{2} = 1$$

$$\Rightarrow \text{slope} = \frac{1}{\sqrt{2} \sin \theta}$$

Which will be equal to the slope of given line $2x - y = 0$

$$\Rightarrow \frac{1}{\sqrt{2} \sin \theta} = 2 \Rightarrow \sin \theta = \frac{1}{2\sqrt{2}}$$

$$\Rightarrow \cos^2 \theta = 1 - \sin^2 \theta = 1 - \frac{1}{8} = \frac{7}{8}$$

$$\Rightarrow \sec^2 \theta = \frac{8}{7}$$

$$\Rightarrow \tan^2 \theta = \frac{8}{7} - 1 = \frac{1}{7}$$

$$\text{Now, } x_1^2 + 5y_1^2 = 4 \sec^2 \theta + 10 \tan^2 \theta$$

$$= 4 \times \frac{8}{7} + 10 \times \frac{1}{7} = \frac{42}{7} = 6$$

Q3

$$E : \frac{x^2}{\cos^2 \theta} + \frac{y^2}{1} = 5 \Rightarrow e_e = \sqrt{1 - \frac{\cos^2 \theta}{1}} = \sin \theta$$

$$H : \frac{x^2}{1} - \frac{y^2}{\cos^2 \theta} = 10 \Rightarrow e_H = \sqrt{1 + \cos^2 \theta}$$

Hyperbola

Hints & Solutions

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Mathematics

$$\therefore e_H = \sqrt{5}e_e \Rightarrow 1 + \cos^2 \theta = 5 \sin^2 \theta \Rightarrow \sin \theta = \frac{1}{\sqrt{3}}$$

$$\begin{aligned}\text{Length of latus rectum of ellipse} &= \frac{2a^2}{b} \\ &= \frac{2(5 \cos^2 \theta)}{\sqrt{5}} \\ &= 2\sqrt{5} \left(\frac{2}{3} \right) = \frac{4\sqrt{5}}{3}\end{aligned}$$

Q4

Here ellipse : $3x^2 + 4y^2 = 12$

$$\Rightarrow \frac{x^2}{4} + \frac{y^2}{3} = 1$$

$$\Rightarrow b^2 = a^2 (1 - e^2)$$

$$\Rightarrow e = \frac{1}{2}$$

$$\text{Foci} = (\pm 1, 0)$$

Now for Hyperbola -

$$2a = \sqrt{2} \Rightarrow a = \frac{1}{\sqrt{2}} \&ae = 1 \Rightarrow e = \sqrt{2} \Rightarrow b = \frac{1}{\sqrt{2}}$$

$$\text{So equation of hyperbola: } \frac{x^2}{\frac{1}{2}} - \frac{y^2}{\frac{1}{2}} = 1$$

$$\Rightarrow x^2 - y^2 = \frac{1}{2}$$

So option (2) does not satisfy it.

Q5

$$\therefore \text{equation of hyperbola is : } \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

$$\therefore \text{ it passes through (3,3): } \frac{1}{a^2} - \frac{1}{b^2} = \frac{1}{9} \dots (1)$$

equation of normal at point (3,3) is :

$$\frac{x-3}{\frac{1}{a^2} \cdot 3} = \frac{y-3}{-\frac{1}{b^2} \cdot 3}$$

$$\therefore \text{ It passes through (9,0): } \frac{6}{\frac{1}{a^2}} = \frac{-3}{\frac{1}{b^2}}$$

$$\therefore \frac{1}{b^2} = \frac{1}{2a^2}$$

From equation (1) and equation (2)

$$a^2 = \frac{9}{2}, b^2 = 9$$

$$\therefore \text{ Eccentricity} = e, \text{ then } e^2 = 1 + \frac{b^2}{a^2} = 3$$

$$\therefore (a^2, e^2) = \left(\frac{9}{2}, 3 \right)$$

Hyperbola

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Mathematics

Q6

General tangent to hyperbola in slope form is

$$y = mx \pm \sqrt{100m^2 - 64}$$

and that of circle is

$$y = mx \pm 6\sqrt{1 + m^2}$$

For common tangent

$$36(1 + m^2) = 100m^2 - 64$$

$$100 = 64m^2$$

$$\Rightarrow m^2 = \frac{100}{64}$$

$$\therefore c^2 = 36 \left(1 + \frac{100}{64} \right) = \frac{164 \times 36}{64} = \frac{369}{4}$$

$$\Rightarrow 4c^2 = 369$$

Q1 JEE Main 2020 - 8 Jan (Evening)

If a hyperbola passes through the point $P(10, 16)$ and it has vertices at $(\pm 6, 0)$, then the equation of the normal to it at P is :

- (A) $2x + 5y = 100$
- (B) $2x + 5y = 10$
- (C) $2x - 5y = 100$
- (D) $5x + 2y = 100$

Q2 JEE Main 2020 - 2 Sep (Morning)

A line parallel to the straight line $2x - y = 0$ is tangent to the hyperbola $\frac{x^2}{4} - \frac{y^2}{2} = 1$ at the point (x_1, y_1) . Then $x_1^2 + 5y_1^2$ is equal to :

- (A) 8
- (B) 6
- (C) 10
- (D) 5

Q3 JEE Main 2020 - 2 Sep (Evening)

For some $\theta \in (0, \frac{\pi}{2})$, if the eccentricity of the hyperbola, $x^2 - y^2 \sec^2 \theta = 10$ is $\sqrt{5}$ times the eccentricity of the ellipse, $x^2 \sec^2 \theta + y^2 = 5$, then the length of the latus rectum of the ellipse, is :

- (A) $2\sqrt{6}$
- (B) $\frac{2\sqrt{5}}{3}$
- (C) $\frac{4\sqrt{5}}{3}$
- (D) $\sqrt{30}$

Q4 JEE Main 2020 - 3 Sep (Morning)

A hyperbola having the transverse axis of length $\sqrt{2}$ has the same foci as that of the ellipse of $3x^2 + 4y^2 = 12$, then this hyperbola does not pass through which of the following points?

- (A) $\left(-\sqrt{\frac{3}{2}}, 1\right)$
- (B) $\left(\sqrt{\frac{3}{2}}, \frac{1}{\sqrt{2}}\right)$

Hyperbola

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Questions with Answer Keys

Mathematics

(C) $\left(\frac{1}{\sqrt{2}}, 0\right)$

(D) $\left(1, -\frac{1}{\sqrt{2}}\right)$

Q5 JEE Main 2020 - 4 Sep (Morning)

Let $P(3, 3)$ be a point on the hyperbola, $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$. If the normal to it at P intersects the x-axis at $(9, 0)$ and e is its eccentricity, then the ordered pair (a^2, e^2) is equal to :

(A) $(9, 3)$

(B) $\left(\frac{9}{2}, 3\right)$

(C) $\left(\frac{3}{2}, 2\right)$

(D) $\left(\frac{9}{2}, 2\right)$

Q6 JEE Main 2020 - 5 Sep (Evening)

If the line $y = mx + c$ is a common tangent to the hyperbola $\frac{x^2}{100} - \frac{y^2}{64} = 1$ and the circle $x^2 + y^2 = 36$, then which one of the following is true?

(A) $5m = 4$

(B) $8m + 5 = 0$

(C) $c^2 = 369$

(D) $4c^2 = 369$

Answer Key

Q1 (A)

Q2 (B)

Q3 (C)

Q4 (B)

Q5 (B)

Q6 (D)